

Effects of IMD on *A. caliginosa*

Protocol

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Objective : Construction of dose-response curves for growth and reproduction.

1 Generalities

1.1 Earthworms

Earthworms for these preliminary tests are *A. caliginosa* individuals from (or bred from individuals) fetch in a meadow in Versailles (Les Closeaux).

1.2 Design

Cosms are made with 200 g of Closeaux soil and earthworms are fed with 3 g/ind/14d.

Each modality is made with 5 replicates for growth and 5 replicates for reproduction.

The growth test lasts 28 days and the reproduction test lasts 28 days with a monitoring of cocoons of 28 days after the test.

For the growth test, one replicate correspond to 1 individuals per cosm. For the reproduction test, one replicate correspond to 2 individuals per cosm.

Table 1.1: Tested concentrations (mg/kg)

Tested concentrations (mg/kg)
0.28
0.50
0.89

We need :

- 20 juveniles for the growth experiment,
- 40 adultes for the growth experiment.

1.3 Soil and horse dung needs

We need :

- 8 kg of Closeaux soil,
- 360 g of horse dung.

2 Protocol

2.1 Calculations

Table 2.1: Soil preparation for each IMD concentration

Concentration (mg/kg)	Q_{IMD} (mg)	Q_{tot} soil (g)	V_{eau} (CRE 70%)	V_{prim} (mL)
0.28	0.616	2200	602.756	3.08
0.50	1.100	2200	602.756	5.50
0.89	1.958	2200	602.756	9.79

18.37 mL of primary solution n°1 are needed (3.674 mg of IMD).

2.2 D-1 - Preparation of chemical, soils and horse dung

Preparation of the primary solution :

- 20 mg of IMD is weighted in a small cup,
- It is introduced in a 100 mL volumetric flask with small volumes of milliQ water,
- The volumetric flask is filled,
- An ultrasound bath is used to completely dissolve IMD (around 20 min).

Preparation of soils : (water used = Volvic)

- For the controls :
 - 2200 g of soil is weighted in an inox recipient,
 - 602.8 mL of water are added (do not mix),
 - The recipient is covered with aluminum.
- For each concentration
 - 1100 g of dry soil is weighted in an inox recipient twice,
 - Recipient 1 is humidified with 301.4 mL of water,
 - For recipient 2 :
 - * The corresponding V_{prim} is used to create a final solution of 100 mL of water,

- * This solution is added drop by drop with a glass pipette on the surface of the soil,
- * The rest of the water (total water added = 301.4 mL) is added on the surface of the soil.
- Recipients are covered with aluminum.

Preparation of horse dung :

- 450 g of dry horse dung are prepared with 1590 mL of water,
- 45×13.6 g of this preparation is weighted in small boxes for the growth cosms,
- 45×27.2 g of this preparation is weighted in small boxes for the reproduction cosms.

2.3 D0 - Preparation of cosms and earthworms introduction

For each concentration :

- Recipient 1 and recipient 2 are mixed individually and then together for at least 10 min,
- This preparation is used to prepare 10 cosms (254.8 g),
- The rest of this preparation is put into sample pots that are kept in a thermoregulated chamber at -20°C ,
- In 5 cosm (growth), 13.6 g of horse dung is added and mixed,
- In 5 cosm (reproduction), 27.2 g of horse dung is added and mixed.

Earthworms introduction :

- For growth, earthworms are weighted individually and put in their respective cosms,
- For reproduction, earthworms are weighted in groups of 2 and put in their respective cosms,
- Cosms are weighted and their weight is written on their labels to be able to follow the soil humidity,
- Cosms are kept in a thermoregulated enclosure at 18°C .

2.4 Maintenance

On D7, D14 and D21, soil humidity is checked.

On D14, earthworms are fed and weighted the same way as D0. Cosms are weighted again.

2.5 D28 - End of exposition

On D28, earthworms are weighted and put on wet filter paper in Petri dishes. Soil samples are taken for each concentration in 125 mL recipients.

2.6 D31 - Freeze of individuals and cocoons recovery

Earthworms are weighted and put in test tubes in a -80°C refrigerator.

2.7 D31-56 - Cocoons

Each day of the working week, cocoons eclosions are checked.